🔥 Practical Cybersecurity Roadmap (Beginner to Advanced)

100% Hands-On | No Theoretical Tests | Focus on Real-World Skills

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Phase 1: Fundamentals (5-6 months)

1️⃣ CompTIA A+ (Networking & Hardware Basics)

Learn These Topics:

- **Networking Fundamentals**: Ports, protocols (TCP/UDP), LAN/WAN concepts
- **Network Devices**: Routers, switches, firewalls, access points
- **Basic Troubleshooting**: Diagnosing network issues, command-line tools (ipconfig, ping, netstat)
- **Hardware & OS Basics**: BIOS/UEFI, file systems, storage, Linux vs. Windows

Watch:

- *Professor Messer's CompTIA A+ Free Course*
- *Mike Meyers' CompTIA A+ (Udemy)*

Hands-on Labs:

- Set up **a home network with a router & VLANs**
- Use **ping, traceroute, and netstat for troubleshooting**
- Install & configure **Windows & Linux on VirtualBox/VMware**

2️⃣ Networking (Computer Networking + Network+)

Learn These Topics:

- **Networking Models**: OSI & TCP/IP layers, encapsulation
- **IP Addressing & Subnetting**: CIDR, subnet masks, IPv4 & IPv6
- **Protocols & Packet Analysis**: DNS, DHCP, ARP, ICMP, Wireshark
- **Wireless & VPN Security**: WPA2/WPA3, VPN tunneling, VLANs

  Wi-Fi Encryption: **WEP, WPA, WPA2, WPA3**
  Wireless Attacks: **Evil twin, deauthentication, MAC spoofing**

VPN Protocols: **PPTP, L2TP/IPSec, OpenVPN, WireGuard**
VLAN Security: **Segmentation, VLAN hopping prevention**

Watch:

- *Computer Networking: A Top-Down Approach* (Read Chapters 1-3)
- *Professor Messer's Network+ Free Course*
- *Mike Meyers' Network+ (Udemy)*

Hands-on Labs:

- Capture & analyze packets with **Wireshark**
- Configure & test **firewalls, VPNs, and VLANs**
- Set up a **home network lab with multiple VMs**

3️⃣ Security Fundamentals (CompTIA Security+)

Learn These Topics:

- **Threats & Attacks**: Malware, phishing, insider threats, DoS
- **Encryption & Cryptography**: AES, RSA, hashing (SHA-256)
- **Access Control & Authentication**: MFA, SAML, OAuth, LDAP
- **Network Security**: Firewalls, IDS/IPS, SIEM, VPNs
- **Endpoint Security**: Antivirus, patching, device hardening

Watch:

- *Professor Messer's Security+ Free Course*
- *Jason Dion's Security+ Bootcamp (Udemy)*

Hands-on Labs:

- Scan a network with **Nmap & Metasploit**
- Perform **MITM attacks using Ettercap**
- Configure **Windows Defender, Linux firewall (iptables)**

4️⃣ Linux & Windows Security

Learn These Topics:

- **Linux CLI & Security**: File system, users, permissions, SSH hardening
- **Windows Security**: Active Directory, Group Policy, event logs
- **Privilege Escalation**: sudo/root in Linux, UAC bypass in Windows
- **Common Exploits**: Misconfigurations, weak credentials, unpatched software

Watch:

- *Linux Command Line Basics (NetworkChuck)*

- *Windows Active Directory Security (Udemy / Pentester Academy)*

Hands-on Labs:

- Secure **SSH, firewall, and sudo rules on Linux**
- Analyze **Windows event logs & Linux syslogs for attacks**

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5 Scripting & Automation (Golang for Security)

Learn These Topics:

- **Golang Basics**: Variables, loops, functions, concurrency (goroutines)
- **Networking in Go**: HTTP requests, sockets, packet sniffing
- **Security Automation**: Writing custom port scanners, vulnerability scanners

Watch:

- *Go for Beginners (NetworkChuck)*
- *Practical Golang for Security Engineers (Udemy)*

Hands-on Labs:

- Write a **port scanner in Go (like Nmap)**
- Automate **log analysis for security monitoring**
- Develop a **brute-force tool with goroutines**

Phase 2: Core Cybersecurity Skills (3-4 months)

6 Web Security & Ethical Hacking

Learn These Topics:

- **Web Security Basics**: HTTP, HTTPS, cookies, sessions
- **OWASP Top 10 Attacks**: SQL Injection, XSS, CSRF, SSRF
- **API Security**: OAuth flaws, JWT attacks, API fuzzing

Watch:

- *Web Security Academy (PortSwigger – Free)*
- *Hack The Box Academy (Beginner Labs)*

Hands-on Labs:

- Exploit **Damn Vulnerable Web App (DVWA)**
- Use **Burp Suite** for request manipulation

7️⃣ Penetration Testing & Red Teaming

Learn These Topics:

- **Pentesting Phases**: Recon, scanning, exploitation, post-exploitation
- **Active Directory Attacks**: Kerberoasting, pass-the-hash, Golden Ticket

Watch:

- *TCM Security Practical Ethical Hacking Course*

Hands-on Labs:

- Hack **Metasploitable 2, TryHackMe, Hack The Box**
- Perform **Active Directory attacks in a lab**

Phase 3: Advanced Cybersecurity Topics (6+ months)

8️⃣ Cryptography & Digital Forensics

Learn These Topics:

- **Encryption Standards**: AES, RSA, ECC, Diffie-Hellman
- **Forensics Process**: Imaging, memory forensics, artifact analysis

Watch:

- *Crypto & Hashing Basics (NetworkChuck)*
- *DFIR Course (SANS)*

Hands-on Labs:

- Crack **hashed passwords using Hashcat**
- Perform **digital forensic analysis using Autopsy**

9️⃣ Security Operations & Incident Response

Learn These Topics:

- **SIEM & Log Analysis**: Splunk, ELK stack, Graylog
- **Malware Analysis**: Sandboxing, static & dynamic analysis
- **Incident Response**: Playbooks, threat hunting

Watch:

- *SOC Analyst Training (John Hammond – YouTube)*

Hands-on Labs:

- Detect **malware in memory dumps using Volatility**

- Analyze **network logs to detect attacks**

Phase 4: Specialization & Real-World Application

✅ **Capture The Flag (CTF) Challenges:**

- **TryHackMe:** “Pre Security” & “Pentester Pathway”
- **Hack The Box:** Beginner to Pro Labs
- **PicoCTF:** Cybersecurity challenges

✅ **Bug Bounty & Pentesting Practice:**

- Join **HackerOne, Bugcrowd**
- Set up a **home security testing lab**

✅ **Certifications (Optional but Recommended):**

- **Beginner:** CompTIA Security+, Network+, A+
- **Intermediate:** eJPT, CEH
- **Advanced:** OSCP, CISSP, GPEN

🔥 Why This Roadmap is Practical?

✅ **80% Hands-on** – Real-world tools, hacking simulations, labs
✅ **No Theoretical Tests** – Only **labs & real-world exercises**
✅ **Includes Golang for Security Automation**
